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TUTORIAL

Theory and Finite-Element Simulation Methodology of Gas Discharge Plasmas

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ABSTRACT This comprehensive research work explores the intricate realm of gas discharge plasma simulations, offering a profound understanding of its theoretical framework and providing a detailed description of the finite-element simulation process in COMSOL Multiphysics software. Firstly, a solid mathematical foundation essential for modeling gas discharge plasmas is established. Subsequently, a detailed finite-element simulation methodology is presented, along with practical guidance on their implementation within the software and an explanation of their theoretical basis. This includes defining the computational domain, setting appropriate initial and boundary conditions, configuring adaptive meshing, addressing computational challenges, and fine-tuning solver settings. Building upon the provided theoretical framework and simulation methodology, a comprehensive case study focusing on negative corona discharges—exemplifying low-temperature plasmas—is presented. First, we provide a comparative parametric study on the influence of various needle electrode geometries and applied voltage magnitudes on the dynamics of negative corona discharges, primarily characterized by Trichel current pulses. Next, the effects of different adaptive meshing strategies, stabilization techniques, time integration methods, and linear system solvers on the convergence, accuracy, and efficiency of the simulations are examined. This paper serves as a valuable resource for students, researchers, engineers, and practitioners engaged in the dynamic field of gas discharge plasma simulations.

INDEX TERMS Finite element simulations, gas discharge plasma, low temperature plasma, negative corona discharge, parametric analysis, simulation methodology, Trichel pulses, tutorial.

I. INTRODUCTION

Electrical gas discharges preceding dielectric breakdowns, such as corona discharges, streamers, dielectric barrier discharges (DBDs), atmospheric-pressure plasma jets, and glow discharges, are characterized by lower temperatures and current densities compared to arcs and sparks. Accordingly, they are classified as low-temperature plasmas; furthermore, they also fall into the subcategories of non-thermal and non-equilibrium plasmas, which are characterized by having electron temperatures significantly higher than the neutral gas temperature. Due to their unique properties,

gas discharge plasmas have become increasingly important in various fields such as materials processing [1], [2], biomedical applications [3], [4], mass spectroscopy [5], [6], environmental remediation [7], [8], and agriculture [9], [10]. Additionally, the study of its behavior plays a crucial role in designing reliable insulation systems for electrical equipment and power system components, since they are predecessors of highly unfavorable breakdowns and flashovers.

Until the late twentieth century, experimental studies were the main source of knowledge about discharge plasmas. However, with technological advances in computers, new avenues for gaining deeper insights into their complex behavior have been opened up. Effective computational models and

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simulations aid in optimizing the design and performance of both plasma-based technologies and electrical insulation systems.

Discharge plasma can be modeled and accordingly modeled by describing the dynamics and interactions of its constituent particles which is achieved by finding a solution to the particles' distribution function, which adheres to Boltzmann's equation. One way to address this is to solve this equation directly in an individual manner to investigate the most probable distribution and probability of deviations from it for each neutral and charged particle. While this direct solution approach offers profound insights into the microscale dynamics of discharge plasma, it requires extensive computational resources and is often infeasible for larger or more complex applications. Concerning this limitation, in a more practical approach, the kinetic (fluid) model provides a framework for a macroscopic scale, based on averaged quantities of particles (Kim et al. [11] review the evolution of this method over time.) These quantities are derived from Boltzmann's equation by taking velocity moments of it. However, since the characteristic time scales for the constituent particles are greater than their involved processes, only the lowest (zero-order) moment, i.e., the conservation of density equation, is typically sufficient to accurately model discharge plasmas [12]. The kinetic model proves particularly effective in scenarios that require detailed temporal dynamics without the granularity of individual particle motion. This attribute is invaluable in designing efficient plasma reactors and is preferred in industries such as semiconductor manufacturing, where understanding general trends and behaviors is more pivotal than mapping exact particle trajectories.

In most cases, to describe the main physics of gas plasma discharges, a common approach is to use conservation of density equations, also known as continuity equations, for three generic kinds of electrically charged particles, namely electrons, positive ions, and negative ions. These equations are typically complemented by Poisson's equation and the resultant is a set of partial differential equations (PDEs) called the hydrodynamic (drift-diffusion) model to simulate the non-thermal, non-equilibrium discharge plasmas.

In this paper, we present a comprehensive description of the hydrodynamic model, in parallel with an in-depth simulation methodology of gas discharge plasmas utilizing COMSOL Multiphysics software. This paper significantly contributes to enhancing both understanding and practical utilization of finite-element simulation techniques for gas discharge plasma phenomena by providing detailed, step-by-step guidance on configuring and implementing these simulations within the software framework, focusing particularly on optimizing the settings for accuracy and efficiency. We then exemplify the theoretical framework and detailed simulation process outlined in this paper through a comprehensive case study. This case study focuses on stable partial breakdowns occurring in gaseous media, commonly referred to as corona discharges. It involves a needle-plane electrode system

subjected to a negative DC voltage, with atmospheric air as the dielectric medium. We conduct comparative studies with parametric and sensitivity analyses to shed light on how key factors influence the characteristics of negative corona discharges, while also assessing the accuracy of the simulation output and the efficiency of solution convergence.

II. HYDRODYNAMIC MODEL AND SIMULATION DETAILS

Implementation of the hydrodynamic model to simulate the non-thermal gas discharges is a complex task as it involves fluid dynamics, plasma transport phenomena, and electric fields. In this regard, COMSOL Multiphysics software has proven to be a valuable resource for researchers which is a powerful finite element method (FEM)-based platform that provides a wide range of modules for modeling various physical phenomena. FEM is a numerical technique used to approximate the solutions of PDEs by dividing the computational domain into finite subdomains and constructing a piecewise polynomial approximation over each subdomain. This software offers a wide range of modules comprising built-in, but customizable equations and interfaces with a wide range of boundary and initial condition options. These are complemented by user-defined computational domain and meshing configuration to model various physical phenomena.

The hydrodynamic model can be implemented using some possible combinations of interfaces. A suitable option is to use a combination of an *Electrostatics (ES)* interface along with three *Transport of Diluted Species (TDS)* interfaces. To effectively capture the initiation and evolution of the discharge plasma over time, next, a *Time Dependent* study must be selected. This method of simulation, the details of which are presented in this paper, effectively encompasses the fundamental physics of the discharge plasma, with a specific emphasis on its electrical characteristics. Alternatively, the *Plasma* interface can be used to simulate gas discharges that involve multiple particles (charged and neutral) and complex reactions. This module allows for the tracking of each species' evolution and influence during the discharge and offers detailed insights into the chemical dynamics of the discharge. Nonetheless, this approach requires significantly more time and computational resources [13]. The following subsections elaborate on the hydrodynamic model and its simulation details.

A. GOVERNING EQUATIONS AND DEFINITION OF MODEL PARAMETERS AND VARIABLES/FUNCTIONS

As mentioned earlier, the hydrodynamic model addresses the kinetics of charged species within a discharge plasma as movements of electronic and ionic fluids. Accordingly, a statistical approach using averaged quantities is employed to describe the collective motion and interactions of these particles. This concludes in a set of PDEs comprising three continuity equations for electrons, positive, and negative ions Eqs. (1)-(3) strongly coupled with Poisson's equation for

local electric field calculation Eqs. (4), (5):

$$\frac{\partial c_e}{\partial t} + \nabla \cdot (-D_e \nabla c_e + \mathbf{u}_e c_e) = R_e \quad (1)$$

$$\frac{\partial c_p}{\partial t} + \nabla \cdot (-D_p \nabla c_p + \mathbf{u}_p c_p) = R_p \quad (2)$$

$$\frac{\partial c_n}{\partial t} + \nabla \cdot (-D_n \nabla c_n + \mathbf{u}_n c_n) = R_n \quad (3)$$

$$\nabla \cdot (\varepsilon_0 \varepsilon_r \nabla V) = -N_A q (-c_e + c_p - c_n) = -\rho_v \quad (4)$$

$$\mathbf{E} = -\nabla V \quad (5)$$

Here, subscripts e , p , and n indicate electron, positive ion, and negative ion species, respectively, and subscript i is used to index them in this paper. In the hydrodynamic model, for each continuity PDE Eqs. (1)-(3), a **TDS** interface must be created. These PDEs imply the charge species' density conservation by representing their dynamics and interactions.

On one hand, their dynamics and fluidity are characterized by their number density, convective, and diffusive flow:

- The charged species' number density is denoted by $n_i = N_A c_i$, m^{-3} , where c_i represents the molar concentration (molarity, or simply, concentration) of the corresponding charged particle, $\text{mol} \cdot \text{m}^{-3}$, and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ is Avogadro constant.
- The convective flux $\mathbf{u}_i c_i$, $\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$, represents the interelectrode drift of charged particles subject to electrostatic force in the electric field $\mathbf{E}$ (es.normE), $\text{V} \cdot \text{m}^{-1}$, where $\mathbf{u}_i = \mu_i \mathbf{E}$ is the drift velocity, $\text{m} \cdot \text{s}^{-1}$, and μ_i is the mobility, $\text{m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$. The velocity vector for each charged species must be entered through the **Transport Properties** node in its components, e.g., r - and z -component in cylindrical coordinates. They can be inputted as $\mu_i \mathbf{E}_r$ and $\mu_i \mathbf{E}_z$ if mobility values are given, and $|\mathbf{u}_i| \mathbf{E}_r / |\mathbf{E}|$ and $|\mathbf{u}_i| \mathbf{E}_z / |\mathbf{E}|$ for given velocity magnitudes. Negatively charged particles' velocity and their components must have a negative value to account for their motion in the opposite direction of the electric field.
- The diffusive flux $\mathbf{J}_i = D_i \nabla c_i$, $\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$, emerges from particles' density gradients, ∇c_i , $\text{mol} \cdot \text{m}^{-4}$, and is characterized by diffusion coefficient, D_i , $\text{m}^2 \cdot \text{s}^{-1}$. It should be emphasized that in each **TDS** interface the convective term must be included and it must be in the conservative form, otherwise, a non-conservative formulation can lead to unphysical behavior and violation of the conservation of particle density.

On the other hand, particles' interactions lead to the generation or loss of charged species and the development of space charge. In the hydrodynamic model, depending on the type of discharge plasma, generic coefficients for rates of essential mechanisms are used, the algebraic sum of which equals the so-called source term, R_i , $\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$, for each type of charged particles:

$$R_e = \alpha c_e |\mathbf{u}_e| - \eta c_e |\mathbf{u}_e| - N_A \beta_{ep} c_e c_p + R_0 + R_{ph} \quad (6)$$

$$R_p = \alpha c_e |\mathbf{u}_e| - N_A \beta_{ep} c_e c_p - N_A \beta_{pn} c_p c_n + R_0 + R_{ph} \quad (7)$$

$$R_n = \eta c_e |\mathbf{u}_e| - N_A \beta_{pn} c_p c_n \quad (8)$$

The following descriptions outline the mechanisms involved in the generation and loss of charged particles:

- When subjected to an external electric field, electrons may acquire adequate energy to undergo inelastic collisions with neutral atoms or molecules, resulting in the production of an additional electron and a positive ion. This phenomenon is known as impact ionization and is characterized by the Townsend coefficient, α , m^{-1} , which quantifies the rate of such ionizing collisions. This process enables the rapid multiplication of charged species and leads to the formation of the conical-shaped electron avalanche with lighter and more rapid electrons diffused at its head and heavier positive ions concentrated at its tail.
- Another process leading to the generation of additional electrons is photoionization; if the energy of an electron in a collision is not enough to accomplish the ionization, it may excite the neutral molecule to a higher energy state. Upon relaxation from this excited state, the molecule emits the energy of a photon which may be enough to ionize another neutral atom or molecule. This process is called photoionization, the rate of which is indicated by R_{ph} , $\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$, and can be effectively computed by the approach presented in [14].
- Another important mechanism is electrons' attachment to neutral, electronegative gas molecules concluding in the loss of an electron and generation of a negative ion. The attachment coefficient is shown by η , m^{-1} , and the term including it specifies the generic rate of this process.
- Recombination mechanisms, the rates of which are denoted by β_{ep} and β_{pn} , $\text{m}^3 \cdot \text{s}^{-1}$, refer to the processes in which two oppositely charged species combine to form a neutral molecule or atom, resulting in a decrease in the number of charged particles in the plasma.
- R_0 , $\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$, is the rate of background ionization due to sources such as cosmic rays and natural radioactivity.

It should be noted that under specific circumstances, additional mechanisms may either become nonnegligible or conversely, further simplifications can be made.

As evident from the equations, velocity and consequently convective flux of charged particles as well as α and η are functions of the local electric field. Therefore, Poisson's equation complements the continuity equations to account for the mutual influence of space charge and interelectrode electric field. In (4) and (5), $\varepsilon_0 = 8.85 \times 10^{-12} \text{ F} \cdot \text{m}^{-1}$ is the vacuum permittivity, ε_r is the relative permittivity or dielectric constant, V is the electric potential, and $q = 1.6 \times 10^{-19} \text{ C}$ is the elementary charge. Furthermore, space charge density denoted by ρ_v , $\text{C} \cdot \text{m}^{-3}$, implies the strong coupling and interdependence between continuity and Poisson's equation.

One of the fundamental quantities characterizing the gas discharge plasma is the current flowing in the external circuit

(interelectrode gap) due to the movement of charged species. To calculate the time-varying value of discharge current, widely recognized Sato's equation [15] or its extension (incorporating time-varying applied voltage) [16] can be used. Two formulations for discharge current calculation are:

$$I_V = \frac{1}{U_a} \left(q \int_V (-n_e \mathbf{u}_e + n_p \mathbf{u}_p - n_n \mathbf{u}_n) \cdot \mathbf{E} dv + \frac{1}{2} \varepsilon_0 \varepsilon_r \frac{d}{dt} \int_V |\mathbf{E}|^2 dv \right) \quad (9)$$

$$I_V = \frac{1}{U_a} \left(q \int_V (-n_e \mathbf{u}_e + n_p \mathbf{u}_p - n_n \mathbf{u}_n) \cdot \mathbf{E}_L dv + \varepsilon_0 \varepsilon_r \int_V \frac{\partial \mathbf{E}_L}{\partial t} \cdot \mathbf{E}_L dv \right) \quad (10)$$

The former formulation involves the Gaussian electric field $\mathbf{E}$, total local electric field due to applied voltage and formed space charge. However, the latter uses the Laplacian electric field $\mathbf{E}_L$, which involves only external static electric field derived from solely applied voltage. In Eqs. (9) and (10), U_a represents applied voltage, V, and integration is carried out over interelectrode gas volume, v . It should be emphasized that diffusive parts are neglected, here. In the case of using a 2D-axisymmetric model in the cylindrical coordinates system, Eq. (10) can be recast as:

$$I_V = \frac{1}{U_a} \iint_S 2\pi r (-n_e \mathbf{u}_e + n_p \mathbf{u}_p - n_n \mathbf{u}_n) \cdot \mathbf{E}_L dr dz. \quad (11)$$

It is important to observe from the dot products in Eqs. (10) and (11) that external circuit current solely arises from the component of charged particle fluxes that is aligned with the Laplacian electric field direction.

Another approach for calculating the discharge current involves performing a surface integration and determining the charge transport between domain boundaries [17]. This can be done by integrating the charged species fluxes through the cathode/anode surface, in addition to the changes in the induced surface charge at the cathode/anode:

$$I_s = \oint_{ca/an} \left(\sum_{i=e,p,n} N_A \mathbf{F}_i|_{norm} \cdot d\mathbf{S} + \frac{d}{dt} \left(\sigma_{ca/an} \cdot \vec{n} \cdot d\mathbf{s} \right) \right) \quad (12)$$

Here, the integration is performed over the cathode or anode surface. The integrand comprises charged species fluxes through the electrode surface, $\mathbf{F}_i|_{norm} = q c_i \mathbf{u}_i|_{norm}$, $\text{C} \cdot \text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$, ($c_i \mathbf{u}_i|_{norm}$ is the normal total flux of each charged particle, $\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) and the time derivative of surface charge density induced at the electrode.

In COMSOL, one may utilize different integration techniques in space [18]. One method involves defining a domain probe, entering the integrand, and plotting the current as a variable name saved for the domain probe. Another method is to define a space or surface integrating operator and define the discharge current using the name saved for it in the **Variables** node. Note that, for discharge current formulae using volume

and surface integration, the operator must be defined for domains and boundaries, respectively. For gas discharge plasma cases, the default integration order of 4 is suitable and provides sufficient accuracy without significantly increasing the computational cost.

All the constants and variables of the model (except built-in ones) must be defined/entered in the software. Quantities like atmospheric conditions (e.g., temperature, humidity, and pressure), secondary electron emission coefficient, and any other ones with a constant value, can be defined in **Global Definitions** → **Parameters**. On the other hand, varying quantities must be defined as variables or functions through **Global Definitions/Component** → **Variables/Functions**. The analytical expression or value of these quantities can also be directly entered in various nodes that necessitate input of the problem configurations.

In order to determine the appropriate magnitude and function shape of the applied voltage, it is necessary to realize the different voltage levels required for the specific type of discharge plasma being studied. For DC voltage, it can be defined as a constant, or equivalently, as an ideal step voltage with an instantaneous transition from zero to its final value. However, to mitigate any abrupt changes in the system's behavior (for some cases) due to a sudden voltage shift, a non-ideal step voltage with a non-zero rise time can be defined. The effects of various magnitudes and rise times of the applied voltage are numerically studied later in this paper.

Some transport and interaction properties of charged species show a strong dependency on the local electric field and environmental conditions. The ideal gas law, valid for most of the common gases within a specific range of pressure and temperature, can be expressed as $P = N k_B T$, where P indicates the gas pressure, Pa, N denotes the number density of gas molecules, m^{-3} , $k_B = 1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$ is the Boltzmann constant, and T indicates the gas temperature, K. Assuming constant values for these environmental conditions, so-called swarm parameters can be expressed as functions of either the local electric field or reduced electric field which is normalized by N [19] or P [20]. However, different datasets, obtained with either empirical evidence or theoretical models, show some inconsistencies, as reviewed by [21]. Table 1 presents the variables, as per widely used swarm parameters of [22] for air at normal atmospheric conditions, which is also validated by [23] and [24]. Here, using the ideal gas law, N is parameterized and $|\mathbf{E}|/N$, Td, is converted into $|\mathbf{E}|$, V/m. In addition, for the sake of ensuring accurate unit consistency, it is advisable to include appropriate unit conversions within the expressions of various quantities.

B. COMPUTATIONAL DOMAIN: GEOMETRY AND MATERIALS

Depending on the intended plasma application or the type of insulation system to be designed, different models can be employed—for either experimentation or simulation—to imitate the characteristics of a real-world electrode

TABLE 1. Variables used in the paper.

Name: Description	Expression/Value
μ_e [$\text{m}^2/(\text{V.s})$]: Electron Mobility	$1.9163E^{-0.25}$
α [$1/\text{m}$]: Ionization coefficient	$3.5 \times 10^5 \exp(-1.65 \times 10^7/E)$
β [$1/\text{m}$]: Attachment Coefficient (2-Body Att.)	$1.5 \times 10^3 \exp(-2.5 \times 10^6/E)$
β_{ep} : Electron-Positive Ion Recombination Coeff.	$2 \times 10^{-13} \text{ m}^3/\text{s}$
β_{pn} : Negative Ion-Positive Ion Recombination Coeff.	$2 \times 10^{-13} \text{ m}^3/\text{s}$
γ : Secondary Electron Emission Coeff.	0.004
D_e : Electron Diffusion Coeff.	$0.18 \text{ m}^2/\text{s}$
D_p : Electron Diffusion Coeff.	$2.8 \times 10^{-6} \text{ m}^2/\text{s}$
D_n : Electron Diffusion Coeff.	$4.3 \times 10^{-6} \text{ m}^2/\text{s}$
μ_p : Positive Ion Mobility	$2.43 \times 10^{-4} \text{ m}^2/(\text{V.s})$
μ_n : Negative Ion Mobility	$2.7 \times 10^{-4} \text{ m}^2/(\text{V.s})$
U_m : Applied Voltage	-5.5 kV
R_{oc} : Needle Tip Radius of Curvature	0.25 mm
G : Interelectrode Gap Distance	3.3 mm

system, study the behavior of a dielectric, and design insulation systems. Below are some examples of power system components insulation [25]:

- Conductor-conductor (parallel cylinders) systems can be used for phase-to-phase insulation system studies/designs.
- Conductor- or rod-plane electrode configuration can be selected to simulate phase-to-tower/ground insulation.
- Point/needle-plane electrode system, with the highest electric stress (field strength) required for breakdown, can be utilized to study/design the insulation for systems containing sharp conductors (e.g., protrusion, edge, brim, and any conductor with a small radius of curvature) in front of a grounded flat (with quite a small curvature) object. Similarly, rod-plane electrode systems are suitable for larger cross-sectional objects.

After identifying the electrode system to be simulated, first, the dimensionality of the model and PDEs should be carefully selected in order to reduce computational costs and runtime as much as possible. This can be achieved by identifying the symmetry axes and opting for the minimum dimensions necessary to accurately simulate the model. This can be applied to the aforementioned examples of electrode systems, in which a 2D-axisymmetric model can be used to represent a 3D geometry that has rotational symmetry about an axis. For that purpose, there is a default boundary condition, called **Axial Symmetry**, allowing one to model only a single cross-sectional plane of the actual 3D component, and then assume that the solution in that plane is representative of the entire structure. This simplifies the simulation, reduces the computational cost, and allows for more efficient computation of solutions for problems with cylindrical symmetry.

It is essential to create the geometry with utmost care to ensure that it accurately represents the physical system being simulated. This is especially important in the case of partial breakdowns, where the strongly non-uniform field distribution is very sensitive to the created geometry. Furthermore, simulation of gas plasma discharges for some electrode systems presumably entails using artificial boundaries to limit the computational domain and reduce the computational cost. However, If the boundaries are too close, the simulation may not accurately capture the physical behavior of the system. Additionally, since a non-uniform mesh is necessary, it may be useful to create a separate domain of any desired shape and size. Later in this paper numerical studies are conducted to emphasize the importance of accurate geometry definition and its influence on the accuracy and quality of simulation results.

C. INITIAL CONDITIONS

At the heart of any PDE lies at least one dependent variable which represents the quantity being modeled and evolves over time and space according to the governing PDE. To fully specify the problem, the values of the dependent variables at the beginning of the simulation, i.e., at $t = 0$ s, or the start of the simulation domain, must be defined through the Initial Values node in each interface. These initial conditions serve as the starting point for the numerical FEM-based solution of our coupled, time-dependent PDEs and determine the behavior of the dependent variables in the absence of any external influences.

It is important to point out that when using FEM, the solution at each time step is used as the initial condition for the next time step. In other words, the initial conditions set at $t = 0$ are used to calculate the solution at the first time step of the solver, and the solution at the first time step becomes the new initial condition for the second time step, and so on. As the simulation progresses, the influence of the initial conditions on the solution becomes smaller and smaller, and eventually, the solution is determined entirely by the boundary conditions and the system's behavior over time. However, it is important to choose appropriate initial conditions that are physically realistic and consistent with the problem being solved, since they can significantly affect the accuracy and stability of the numerical solution.

As stated previously, three generic types of charged species' molar concentration c_i are dependent variables for three continuity PDEs in each TDS interface. Usually, at the start of the simulation (prior to applying voltage), the gas is assumed to be electrically neutral, meaning that the quantity of positive and negative charges in the system is balanced. Accordingly, the initial molar concentrations of electrons and positive ions are set to a common value c_0 , while the initial concentration of negative ions is considered to be zero. c_0 , which represents the background density of charged particles in the system, essentially incorporates the spatial distribution of these particles at the start of the simulation (using reserved spatial coordinates variables, e.g., r and z

for 2D-axisymmetric geometry with a cylindrical coordinate system). This distribution can be uniform or non-uniform depending on the physical system being modeled and the assumptions made by the user; for instance, in simulations involving concentric spheres or coaxial cylinders, a uniform distribution of charged species is typically used as the initial condition, whereas for systems like rod-plane or point-plane, where the electric field is more intense near the tip of the electrode, a non-uniform distribution with a greater concentration of charged species near the tip may be used. This helps to capture the non-uniformity of the electric field and provides a more accurate representation of the system.

On the other hand, electric potential V is the dependent variable for the ES interface, and similarly, an initial value, expressing its initial spatial distribution must be assigned to it. This can be done by either defining an analytical expression for the initial spatial distribution of V or by utilizing an actual distribution obtained from a stationary study with only the ES interface. The procedure for the latter is similar to that of achieving the Laplacian field, as previously discussed.

D. BOUNDARY CONDITIONS

In order to achieve a precise solution for a hydrodynamic model using FEM, it is essential to define appropriate boundary conditions as they play a vital role in determining the behavior of the plasma at the region of interest. Boundary conditions are fundamental in modeling as they set the physical limits and behaviors at the edges of the domain, effectively guiding the plasma dynamics and interactions at these interfaces. Incorrect or poorly specified boundary conditions can lead to several issues, including non-convergence of the simulation or an inaccurate depiction of the plasma's behavior. This can result in simulations that fail to reflect the true physical scenarios being modeled, leading to misleading or erroneous results.

Boundary conditions refer to the specifications of the values of the dependent variables (concentration of charged species and electric potential), their derivatives (fluxes and fields), or combinations of these quantities on the physical and artificial boundaries of the computational domain—these three types are known as Dirichlet, Neumann, and Robin boundary conditions, respectively. As stated earlier, in COMSOL, the Axial Symmetry node is active by default on all symmetry axis boundaries. The discussion that follows will cover other boundary conditions for *TDS* and *ES* interfaces individually.

- For exterior boundaries in *TDS* interfaces, by default, the **No Flux** node is the prescribed boundary condition, although other boundary conditions can override it. When the box for convection is checked, the no flux condition is equivalent to setting $-\mathbf{n} \cdot (-D_i \nabla c_i + \mathbf{u} c_i) = 0$, where $\mathbf{n}$ represents the normal vector that points outward from the boundary. The rule of thumb to determine the boundary conditions for continuity PDEs in *TDS* interfaces is that if a positive/negative potential

is applied to an electrode, positively/negatively charged particles are subject to **No Flux** boundary condition, while negatively/positively charged particles have **Outflow** boundary condition (i.e., they have solely convective flux and diffusive flux is zero, $\mathbf{n} \cdot D_i \nabla c_i = 0$); conversely, the grounded electrode located in front of it has an **Outflow** boundary condition for negatively/positively charged particles and a **No Flux** for positively/negatively charged particles. Additionally, a **Flux** boundary condition can be used to model any additional source of charged particles originating from electrodes. If artificial boundaries are far enough from the main discharge region, they can be assigned an **Outflow** boundary condition.

- In the *ES* interface, the default boundary condition at exterior boundaries is **Zero Charge**. This condition is defined mathematically as $\mathbf{n} \cdot \mathbf{D} = 0$, where $\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}$ is the electric displacement vector, $\text{C} \cdot \text{m}^{-2}$. This condition ensures that the normal component of the electric field is continuous across the boundary (this is not the case for any interior boundary), which is physically meaningful since there can be no accumulation of charge on a boundary with no net charge. The **Zero Charge** boundary condition can be applied to artificial boundaries that are sufficiently far from the computational domain. However, on the surface of electrodes with specific electric potential, this condition must be overridden. As detailed earlier, this can be done by applying a constant or analytical expression of electric potential through the **Electric Potential** node ($V_0 = U_a$), or by setting the potential to zero ($V = 0$) using the **Ground** node.

E. MESHING CONFIGURATION

Meshing is the foundation on which FEM stands, where it discretizes the computational domain into a mesh of finite elements, enabling the numerical solution of the PDEs governing the plasma behavior. The quality and resolution of the mesh play a crucial role in the convergence, accuracy, and efficiency of the simulation. A well-designed mesh should adapt to the geometry and physics of the problem, with higher resolution in regions of interest and lower resolution elsewhere.

For example, in a parallel plate system with protrusions, bumps, or ridges, it is necessary to use a denser mesh near these irregularities because the electric field is more non-uniform in these regions. Similarly, in a DBD, the mesh resolution needs to be higher near the dielectric layers to capture the surface charge accumulation and the discharge dynamics. In another notable example, electrode configurations such as rod-plane or needle-plane, with extremely non-uniform interelectrode electric fields, higher mesh resolution is needed in the vicinity of the rod or needle tip to capture the strong field gradients accurately. On the other hand, electrodes with relatively smooth and uniform geometries may not require a high resolution and can be

meshed with coarser elements to conserve computational resources.

However, in some scenarios where plasma evolution covers extensive areas, relying solely on adaptive meshing with a fixed preset mesh configuration may require excessively large area with fine mesh to capture the main plasma activities. This can lead to significant increases in both time and computational resource consumption. In such cases, employing self-aware dynamic mesh refinement techniques, which generate time-variant mesh configuration on demand, proves to be the most efficient approach. A notable example of this can be found in the work by Ghosh and Chaudhury [26], where a dynamic mesh refinement technique is utilized to efficiently simulate the plasma dynamics associated with high-power microwave breakdown, leading to the formation of filamentary plasma structures. This approach not only adapts the mesh size dynamically based on plasma profile changes but also significantly reduces computational overhead by focusing computational effort where it is most needed.

Another important consideration when configuring the meshing is the desired level of visualization quality. Opting for a lower meshing resolution can lead to pixelation in graphics and visualization, resulting in a visual distortion marked by the presence of noticeable pixel elements that can significantly compromise the visual clarity and precision of the graphical representation.

One method of achieving adaptive meshing in COMSOL involves setting *Element Size Parameters*. These parameters, such as *Maximum/Minimum Element Size* and *Maximum Element Growth Rate*, can be specified for particular domains or boundaries (e.g., the domain formed through *Object Type: Curve*, as stated in subsection II-B) in the computational domain. In the following, an instance of adaptive meshing, revealing the sensitivity of the results to varying mesh configurations is presented.

F. COMPUTATIONAL AND INSTABILITY CHALLENGES

The continuity equations are crucial components of the hydrodynamic model for gas discharges to ensure the conservation of mass during simulation. However, numerical simulations of the continuity equation can be challenging due to the presence of high spatial gradients and discontinuities in the plasma density and velocity fields. In other words, the values of these fields change rapidly in a small region of space, which can occur near the plasma sheath or in regions of strong plasma turbulence. These rapid changes can lead to high values of the convective and diffusive fluxes of charged particles, which are proportional to the plasma density and its gradient. When the gradients are very high, the convective flux can dominate over the diffusive flux (i.e., $u_i c_i \gg D_i \nabla c_i$), resulting in very high Péclet numbers, which is a dimensionless number that characterizes the relative magnitudes of convective and diffusive fluxes in a fluid. This can lead to numerical instabilities and errors that need to be addressed through numerical stabilization

techniques. These techniques usually involve either adding artificial diffusion to the equations to dissipate numerical errors and dampen the numerical oscillations in the solution or limiting the convective flux to prevent it from becoming too large. Adopting the former technique, COMSOL provides two general types of stabilization techniques found in *TDS* interfaces:

- **Inconsistent Stabilization (Isotropic Diffusion)**: this stabilization method adds a diffusion term (δ_{id}) equally in all directions, which may not be optimal for flows with anisotropic behavior.
- **Consistent Stabilization**: This stabilization techniques add anisotropic diffusion terms to the continuity equations which are consistent with the underlying physics of the problem. Two types of consistent stabilization are:
 1. **Streamline diffusion** adds artificial terms that are aligned with the direction of the flow (drift velocity), and proportional to the length of the streamline and the gradient of the solution along the streamline.
 2. **Crosswind diffusion** adds artificial diffusion that is perpendicular to the flow direction and has two available options: 1) *Do Carmo and Galeão* and 2) *Codina*. The former (default option), through a geometric interpretation, assumes that the convective velocity is aligned with the gradient of the solution, and the diffusion coefficients are defined accordingly. The latter uses a flux limiter function to limit the magnitude of the artificial diffusion in order to prevent it from becoming too large in regions of high gradients. The former formulation adds more diffusion, is more effective in minimizing under- and overshoots, and works better for anisotropic meshes. However, the latter results in non-linear equations that are likelier to converge.

Overall, the results obtained from the *Streamline diffusion* technique are relatively good, and the computational time required for simulations is reasonable for the test problems. However, the *Crosswind diffusion* technique offers better accuracy for the test systems, albeit at the expense of more computation time. Section III of this paper will involve a numerical comparative investigation of these stabilization techniques, both individually and in combination.

Another challenge in solving the hydrodynamic model is dealing with the large range of spatial and temporal scales involved, spanning from the microscopic scales of individual particles and collisions to the macroscopic scales of the plasma bulk and electrode system dimensions. Furthermore, the strong interdependence and coupling between continuity equations and Poisson's equation results in a highly non-linear system. In numerical methods, non-linearity can lead to convergence issues, such as oscillations, and can increase the complexity of the algorithms required to solve the model. To address these challenges, an effective approach is to utilize

the adaptive meshing technique with increased resolution in regions of interest and to select an appropriate solver and solver settings, as discussed in the following subsection.

G. SOLVER SELECTION AND SETTINGS

Discretizing the computational domain by building a mesh allows the linear system solvers to handle the linearized versions of the non-linear equations and approximate the solution of the governing equations—for their corresponding dependent variable—at discrete points within each element. Depending on the model and modules/interfaces used, COMSOL enables using direct, iterative, or multigrid system solvers. For gas discharge plasma simulation using *ES* and *TDS* interfaces, it provides four direct linear system solver options of *MUMPS* (default), *PARDISO*, *SPOOLES*, and *Dense matrix*, which compute the solution by direct factorization methods. Solver settings and selection can be performed through the *Time-Dependent Solver* node, and its sub-node, *Direct*, respectively. An alternative is to employ *Iterative* linear system solvers. This method involves iteratively refining an initial guess to the solution and proves particularly efficient when the memory capacity is constrained.

During a time-dependent simulation, time steps are taken by the solver to advance the solution in time. The time step size depends on the stability criteria of the chosen solver and the solution method used. Generally, the smaller the time step size, the more accurate the solution, but at the expense of longer computation times. To address dynamic, time-dependent problems, there exist two primary classifications of *Time Stepping Methods: Implicit* and *Explicit*. *Implicit* methods employ algorithms that derive solutions for the subsequent state by resolving equations that encompass both the system's existing state and a subsequent state. Conversely, *Explicit* methods calculate the system's state at a later temporal instance solely from its current state. The choice between these methods should be made considering the computational characteristics and the trade-offs between stability, accuracy, simulation runtime, and memory usage.

Output times represent the time instances at which the solver generates the solution data and can be specified through either the *Study Settings* tab or in the *General* tab within the *Time-Dependent Solver* node. These times can be specified by the user and may or may not coincide with the *Time steps taken by solver*.

The Backward Differentiation Formula (*BDF*) is the default time integration method used by COMSOL to solve time-dependent problems, with the option to change it under *Time-Dependent Solver* → *Time Stepping* tab. *BDF*, as an *Implicit* time-stepping method, approximates the derivatives of the solution at each time step using a weighted sum of the solution values at the previous time steps. This method is particularly effective in capturing the wide range of time scales involved in gas discharge plasma modeling. The choice of *Maximum/Minimum BDF order* should be made based on a trade-off between accuracy and efficiency; such

that higher-order *BDF* methods require more previous time steps to calculate the derivative at the current time step, but they typically provide better accuracy and stability. The default values considered by the software take into account several factors in determining them, making them reliable in most cases. Note that, if *Manual* is chosen for *Steps taken by solver*, a constant *BDF order* in the range of 1-5 can be applied. Using lower *BDF orders* is effective in maintaining stability; however, this comes at the cost of significantly damping high frequencies, potentially resulting in outcomes that deviate from reality. Therefore, opting for a variable *BDF order* within a specified range could be a more suitable approach. Also, if desired, the default limit on the acceptable relative/absolute error or discrepancy between the true solution and the computed approximation can be adjusted by specifying *Relative/Absolute Tolerance*. The efficiency and precision of different solvers under various settings are numerically investigated in the second installment of this paper.

III. A COMPREHENSIVE PARAMETRIC STUDY ON NEGATIVE CORONA

In the first part of this paper, Section II, we laid the theoretical foundation for understanding gas discharge plasmas and provided the details of mathematical model and finite-element simulation of these phenomena. We offered a circumstantial description of the hydrodynamic model, which treats charge particle ensembles as fluids and describes their motion and interactions under applied voltage via fluid dynamics principles. In parallel, we detailed a comprehensive simulation methodology within the COMSOL Multiphysics software framework. This encompassed aspects such as defining quantities, delineating the computational domain and geometry, setting proper initial and boundary conditions, configuring adaptive meshing, addressing computational challenges, explaining stabilization techniques, and selecting appropriate solver settings.

In this section, our second installment concentrates on a specific type of low-temperature discharge plasmas – corona discharges under applied negative DC voltage. We aim to exemplify the simulation methodology and principles outlined in Section I through a detailed case study, shedding light on the intricacies of negative corona discharges. When a sharp electrode, like a needle, is subjected to negative DC voltage—within a magnitude range slightly above corona inception voltage—the resulting discharge current flowing between two electrodes starts to exhibit pulsating behavior. This current pulse train, recognized as Trichel pulses, stands out as the most distinctive feature of negative corona discharges, with its magnitude and temporal characteristics influenced by various factors.

Earlier paper, we emphasized the critical role of computational studies in gaining profound understanding of gas discharge plasmas, optimizing various plasma-based applications, and designing reliable insulation systems for electrical equipment. In the context of computational studies

of negative corona discharges, Morrow [27] is recognized for introducing the initial 1D numerical model that integrates Flux-Corrected Transport (FCT) and Finite Difference Methods (FDM), effectively capturing the characteristics of the first Trichel pulse in O_2 . A quasi-1D numerical model was developed by Napartovich et al. [28] to simulate negative corona discharges in air, describing the properties and processes of Trichel pulses using simulation, especially for gaps smaller than 1 cm. Employing a 3D model—reducible to a 2D-axisymmetric model—Akishev et al. [29] expanded on [28] by further examining Trichel pulse production in a needle-plane configuration. Their approach allowed for an anticipation of the current density and charged particles distributions on the cathode surface throughout the phases of discharge current pulsation.

Sattari et al., in [30] and [31], utilized 2D-axisymmetric models using one and three types of charged species, respectively. They employed a hybrid approach to solving the problem, using the Finite-Element Method (FEM) for local electric field determination and FCT method for space charge density calculation. Additionally, in collaboration with Gallo, they applied the FCT method for a parametric analysis, investigating the impacts of the ions mobility, secondary electron emission rate, and the resistance of external circuit on Trichel pulse characteristics [32]. Tran et al. [17] carried out a significant scrutiny on negative air, offering a microscopic perspective on discharge current pulsation under conditions with and without dielectric barriers. Dordizadeh et al. examined the formation of Trichel pulses [33] and investigated the impact of photoionization on the characteristics of Trichel pulse train [34]. In a prior investigation [35], we closely analyzed the initial phases of negative discharge formation, achieved through the application of a 2D-axisymmetric hydrodynamic model. Using the hydrodynamic model, [36] introduced and explored different sets of analytical expressions for ionic reaction rate coefficients. There exists a substantial body of literature that has engaged in both experimental and computational parametric and sensitivity analyses, examining various characteristics of negative corona discharges and their responses to changes in various factors.

Some investigations focus on understanding the influence of changes in electrode system configuration, encompassing variations in needle cathode radii [37], [38], [39] and interelectrode gap distances [37], [40], [41]. Further studies explore negative corona discharges in setups beyond the typical needle-plane electrode system [42], [43], [44], [45], while a considerable volume of research is dedicated to understanding the effects of different air dielectric conditions, such as humidity [46], [47], [48], temperature [49], [50], [51], and pressure [47], [51]. This section of the paper conducts comparative, parametric, and sensitivity analyses on negative corona discharges, aiming to understand how simulation outputs respond to and are influenced by diverse parameters, conditions, and settings. Employing the simulation methodology elucidated in Section II, our primary goal is to uncover the intricate dynamics underlying negative corona

discharges, with a specific focus on Trichel current pulses. Our numerical study is situated within the framework of the needle-plane electrode system subjected to negative DC voltage, with atmospheric air as the dielectric medium. After introducing a base case study, this investigation scrutinizes the impact of needle electrode geometry, applied voltage, meshing configuration, stabilization technique, linear system solver, and time integration method.

A. CASE STUDY OVERVIEW

In this paper, we present a comprehensive case study that builds upon the simulation framework discussed in the previous section. The aim of this case study is to examine the behavior of air discharge plasma in a needle-plane electrode system under normal atmospheric conditions, with a negative DC voltage applied. The choice of this particular electrode system is its possibility of being simulated as a 2D-axisymmetric model and capability to generate a strongly non-uniform interelectrode electric field (typically with a Schwaiger factor below 0.25 [52]) when an applied high voltage is introduced. The presence of a sharp potential gradient drop at the electrode tip prevents the avalanches formed within non-uniform fields from extending deeper into the gap towards the opposite electrode. Therefore, in non-uniform fields—unlike uniform and quasi-uniform (weakly non-uniform) fields—a range of luminous and audible discharges can be witnessed much below the breakdown voltage. In gaseous media, these localized partial breakdown phenomena occurring in regions of intense electric fields are commonly referred to as coronas.

The choice of applying a DC voltage with negative polarity is primarily motivated by its ability to induce Trichel current pulses. By exploring the main characteristics of Trichel pulses under various settings, we can gain valuable insights into the dynamics of plasma and investigate the influence of different factors discussed in the preceding section. The numerical simulations in this case study are structured to explore the behavior of gas discharge plasma comprehensively under various conditions, settings, and parameters. Initially, a base case is established, incorporating specifications and settings that are deemed most suitable for the study objectives and serving as a benchmark. Subsequently, to investigate the influence of different factors on plasma behavior, multiple additional cases are systematically conducted. These cases intentionally modify various aspects deviating from the corresponding values in the base case. The sensitivity of simulation outputs to these deliberate deviations is thoroughly evaluated and results are compared against both the base case and each other. The variables employed in all cases are documented in Table 1 unless otherwise specified in the given context.

The simulations were conducted on a workstation featuring an AMD Ryzen Threadripper PRO 5945WX processor with 12 cores, operating at 4.10 GHz, and equipped with 128 GB of usable RAM. The simulation software utilized for these experiments was COMSOL Multiphysics version 6.2. To ensure accurate and meaningful runtime comparisons

between different cases, simulations were conducted sequentially, one at a time. This approach allows for the dedicated allocation of computational resources to each simulation, ensuring a fair and reliable assessment of simulation times for varying conditions and settings. In Subsections III-B and III-C, although we have methodically explored the effects of different needle cathode geometries and applied voltage magnitudes through individual parameter analyses, it is worth noting that we could have alternatively utilized the *Parametric Sweep* study in COMSOL. This function enables the simultaneous examination and comparison of solutions across multiple parameter combinations within a single comprehensive study, proving highly useful for the optimization of parameter settings.

We have rendered all graphs in a vector-based format, assuring clear images that remain sharp even when scaled. The components of the software, such as its modules, interfaces, nodes, and tabs, are written in bold and italic typeface. It is essential to highlight that some of the analyses and conclusions are specific to negative corona discharges and may not be universally applicable to all discharge plasmas. This case study serves as an illustrative example of how to improve our simulations and numerical investigations; it demonstrates the methodology for evaluating various parameters, conditions, and settings to identify optimal simulation inputs tailored to the specific characteristics of the discharge plasma being studied.

B. BASE CASE SPECIFICATIONS

This subsection presents the details of the base case. This setup has undergone robust validation and extensive examination in [17], [38], [41], and [53], instilling confidence in its suitability for addressing the current paper objectives. By presenting the base case item-by-item, the groundwork for further analysis and comparison with subsequent cases featuring modified parameters, conditions, and settings is laid.

1) ELECTRODES GEOMETRY AND COMPUTATIONAL DOMAIN

As previously mentioned in Subsection II-B, needle-plane electrode configurations can be effectively modeled in a 2D-axisymmetric framework. This model adopts cylindrical coordinates as the spatial reference system, where the azimuthal angle φ remains constant, and geometry is represented in the $r - z$ plane. In the context of our base case study, we employ a standardized hyperboloid-shaped needle electrode. In mathematical terms, the equation that describes a vertical hyperbola can be written as:

$$(z - k_h)^2 / b_h^2 - (r - h_h)^2 / a_h^2 = 1 \quad (13)$$

In this equation, (h_h, k_h) , $(h_h, k_h \pm b_h)$, and $(h_h \pm a_h, k_h)$ are coordinates of the center, vertices, and co-vertices of the hyperbola, respectively. Our standard hyperbolic electrode is characterized by positioning the center of the vertical

hyperbola at the origin (0,0). Assuming that the plane electrode is consistently situated at $z = 0$, this configuration results in the vertical hyperbola's semi-major axis being equal to the gap distance between the electrodes, i.e., $b_h = G$. To calculate the radius of curvature of a curve at a specific point, the following equation can be employed.

$$R = \left(1 + (dz/dr)^2\right)^{3/2} / |d^2z/dr^2| \quad (14)$$

Using this equation at the vertex of the standard hyperbola, which corresponds to the needle tip (where $z' = 0$ and $z'' = b_h^2/(a_h^2 z) - z'^2/z$), yields $R = a_h^2/b_h$. When this value is integrated into Eq. (13), the following equation defining the upper branch of the vertical hyperbola is obtained:

$$z = G\sqrt{1 + r^2/RG} \quad (15)$$

This process allows us to precisely define the unique geometry of the standard hyperbolic needle configuration, which is entirely defined by the radius of curvature at the needle tip and the gap distance between the electrodes.

As studied and validated in multiple works ([17], [41], and [53], among others), this paper consistently adopts values of $R = 0.25$ mm and $G = 3.3$ mm across all cases. However, the precise definition of a unique conic cross-sectional needle electrode geometry necessitates the incorporation of an additional parameter beyond the combination of geometric shape type and radius of curvature—a facet often overlooked in existing literature. For instance, in the case of our standard hyperbolic needle, the assumption that the semi-major axis is equal to the gap distance, as defined above, is one such consideration. Yet another accompanying parameter to the radius of curvature is the eccentricity value of the conic section. The influence of this parameter, under a constant radius of curvature, is explored in this paper.

To define the 2D shape of the hyperboloid-shaped needle electrodes, one approach is to utilize a *Parametric Curve* tool found under *Geometry* → *More Primitives*, as illustrated in Figure 1 (a). The computational domain, across all cases, is confined by two artificial boundaries positioned at distances of 6.7 mm and 10 mm from the lateral and upper sides of the needle tip, respectively, forming a half-computational domain measuring 10×10 mm². The computational domain for the base case in the 2D-axisymmetric setting is illustrated in Figure 2.

2) APPLIED VOLTAGE

The minimum voltage required for measurable ionization and corona discharges is called corona onset (inception) voltage, which exhibits negligible dependence on the specific gap length and needle tip radius [54]. In the case of applying negative DC voltage to a needle, just below the corona onset threshold, an irregular pattern of discharge current pulses occurs between the needle and the grounded plane. However, by slightly increasing it slightly above the corona onset voltage a very regular Trichel pulse train

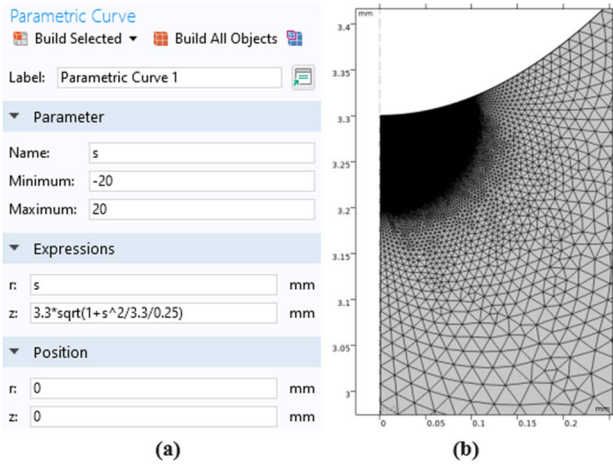


FIGURE 1. (a) Definition of the standard hyperbolic needle geometry (base case) (b) Adaptive meshing configuration around the needle tip.

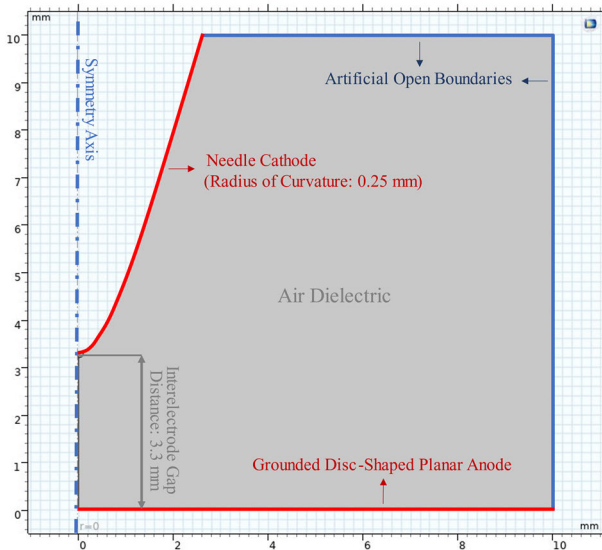


FIGURE 2. Computational domain in 2D-axisymmetric for the base case.

can be achieved, persisting over a wide range of applied voltage, until transitioning into glow discharges. On further increasing the voltage, the production of charged species intensifies, causing the glow discharges to become unstable and eventually resulting in a spark breakdown of the gaseous dielectric. Hence, the selection of the voltage should fall within the range encompassing the onset voltage of Trichel pulses (slightly higher than the corona onset voltage) and its transitional voltage to glow discharge. In the base case, a DC voltage of -5.5 kV is applied using a non-ideal step function with 1 ns rise time.

3) INITIAL AND BOUNDARY CONDITIONS

As stated in Subsection II-C, the simulation assumes the inter-electrode gas is initially neutral, with equal concentrations of electrons and positive ions, while the concentration of

negative ions is set to zero. To enhance the representation of the electric field's non-uniformity and expedite solution convergence, it is assumed that electrons and positive ions obey a Gaussian distribution as follows:

$$c_0 = c_{0,max} \times \exp \left(- (r - r_0)^2 / 2s_0^2 - (z - z_0)^2 / 2s_0^2 \right) \quad (16)$$

Here, $c_{0,max} = 10^{16} \text{ mol} \cdot \text{m}^{-3}$ is the peak initial molar concentration at the needle tip $(r_0, z_0) = (0, G) \text{ mm}$, and $s_0 = 0.025 \text{ mm}$ is the radius of initial distribution. Additionally, a negative spatial distribution of $V_0 = -10^{-5}(r - z)$ is assumed for electric potential.

The boundary conditions are defined on each boundary of the computational domain, ensuring a comprehensive representation of the system's behavior in our simulations, as detailed in [17]. Among these conditions, secondary electron emission—due to the impact of positive ions with the cathode surface—emerges as a crucial mechanism that sustains the negative corona discharge. It can be modeled as an electron flux (in **Electron TDS** $\rightarrow$ **Flux**) equal to $\Gamma_e = \gamma c_p |\mathbf{u}_p|$, $\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$, where γ is a dimensionless number between 10^{-3} and 10^{-2} .

Furthermore, as described in Subsection II-B, when the electrodes do not confine the computational domain, it is necessary to define it using artificial boundaries, often referred to as open boundaries. These boundaries are strategically placed at an optimal distance from the region where main plasma activities occur. This distance is critical: if too close, the boundaries may adversely affect the simulation by artificially constraining the behavior of the plasma; if too far, they unnecessarily increase both computational load and simulation time. In scenarios where the boundaries are sufficiently distant from the primary discharge area, as in the current case, **Outflow** (for **TDS** interfaces) and **Zero Charge** (for **ES** interface) boundary conditions are applied. The latter sets the field component normal to the boundaries to zero, effectively capturing the actual behavior of field and particle dynamics. These conditions permit particles and energy to freely exit the computational domain, mimicking an open environment and ensuring that the simulation accurately represents the actual physical setup.

4) MESHING CONFIGURATION

Following the discourse in Subsection II-E, an adaptive mesh tailored to needle-plane electrode systems, featuring finer mesh size near the needle tip and coarser mesh elsewhere, is crucial for effectively capturing the primary plasma dynamics. This approach ensures efficient utilization of resources, both in terms of memory and time. In the base case, a circular curve (**Object Type: Curve**) with a radius of 0.1 mm is drawn, centered precisely at the needle's tip, to be networked into mesh elements of finer size. Within this domain, the **Element size** is **Customized**, having a **Maximum element size** of 0.001 mm, while its remaining settings are by default configured as **From sequence**. The other sections

of the computational domain employ a **Predefined**, coarser **Element size** categorized as **Extremely fine**. This mesh configuration is visually represented in Figure 1 (b).

5) STABILIZATION TECHNIQUE

In accordance with the discussion presented in Subsection II-F, an imperative measure for ensuring numerical stability and promoting enhanced convergence involves the incorporation of artificial diffusion within the continuity PDEs. In the base case of this study, the selected approach for achieving this stabilization is the integration of anisotropic **Streamline diffusion**, which will undergo comprehensive evaluation and comparison against alternative techniques.

C. IMPACT OF DIFFERENT NEEDLE CATHODE GEOMETRIES

In this section, we narrow our focus to investigate the precise body shape of needle cathodes, all sharing a common radius of curvature of $R = 0.25$ mm and gap distance of $G = 3.3$ mm. Accordingly, this paper comparatively explores how different needle cathodes influence the discharge current characteristics. The analysis, also, involves the quantification of interelectrode electric field uniformity through the Schwaiger factor η , calculated as the ratio of the peak mean to the maximum interelectrode electric field intensities. In the following, first, these needle electrodes are mathematically introduced. This includes needle electrodes with profiles shaped as three conic section types: hyperboloid, paraboloid, and ellipsoid. Additionally, we examine how simulation outputs differ when approximating such electrodes with tangent-circular needles.

1) HYPERBOLOID-SHAPED NEEDLES

Following the mathematical expression provided in (1) and considering the inclusion of the radius of curvature at the needle tip $(0, G)$ equal to a_h^2/b_h , the general description of hyperbolic needles can be expressed as follows:

$$z = b_h \left(\sqrt{1 + r^2 / b_h R} - 1 \right) + G \quad (17)$$

These needles with the given radius of curvature can be characterized by eccentricity rather than their semi-major axis. The eccentricity of a hyperbola, possessing a value greater than 1, can be determined as $e_h = \sqrt{1 + a_h^2/b_h^2} = \sqrt{1 + R/b_h}$.

2) PARABOLOID-SHAPED NEEDLE

The solitary parabolic needle, characterized by an eccentricity value of 1, can be mathematically expressed as follows:

$$z = a_p(r - h_p)^2 + k_p \quad (18)$$

Here, (h_p, k_p) represents the vertex of the parabola. For the parabolic needle, where its axis symmetrically aligns with $r = 0$ and the vertex is located at a distance G from

the grounded plane ($z = 0$), the vertex coordinate (h_p, k_p) corresponds to $(0, G)$. Applying Eq. (14) to compute the radius of curvature at the parabola's vertex: $R = 1/2a_p$. Consequently, this results in the parabolic equation that describes the shape of the needle electrode cross-section along its axis of symmetry as follows:

$$z = r^2/2R + G \quad (19)$$

Similarly, this needle shape can be defined using the **Parametric Curve** tool.

3) ELLIPSOID-SHAPED NEEDLES

These needles are defined by the following mathematical expression:

$$(z - k_e)^2 / b_e^2 + (r - h_e)^2 / a_e^2 = 1 \quad (20)$$

In this equation, (h_e, k_e) , $(h_e, k_e \pm b_e)$, and $(h_e \pm a_e, k_e)$ correspond to the coordinates of the center, vertices, and co-vertices of the ellipse, respectively. When considering vertically oriented elliptic needles, it is necessary for b_e to be greater than a_e . The radius of curvature at the lower vertex $(0, G)$, as determined by Eq. (14), is given by $R = a_e^2/b_e$. Using this value in Eq. (20), the geometry of the elliptic needle can be represented as follows:

$$(z - (b_e + G))^2 / b_e^2 + r^2 / b_e R = 1 \quad (21)$$

Similar to hyperbolic needles, an elliptic needle with a specified radius of curvature can be characterized by its eccentricity value of $e_h = \sqrt{1 - a_e^2/b_e^2} = \sqrt{1 - R/b_e}$, which is always less than 1.

4) TANGENT-CIRCULAR NEEDLES

This paper examines an additional category of needle electrode configurations, referred to as tangent-circular needles [30], which serve as approximations to the conic cross-sectional counterparts—the generic shape of a needle. These needles have cross-sections formed by a circle and its two tangents. In all cases studied in this paper, the circle that forms the basis of the tangent-circular needle electrodes has a consistent radius of $R = 0.25$ mm. These needles are characterized by the angle created when two tangents extend and meet along the axis of symmetry at $r = 0$. By modifying this angle, we can achieve diverse orientations and variations in the electrode geometry.

In this comparative analysis, we investigate cathode needle shapes, including conic cross-sectional needles characterized by varying eccentricity values ranging from 0.975 to 1.120 in increments of 0.005, and tangent-circular needles with angles ranging from 40° to 75° in 5° increments. η , the measure of electric field uniformity, exhibits direct proportionality with eccentricity and angle values across the investigated domains, ranging from 13.3% to 16.0% and 12.1% to 15.5%, respectively, with a subtle departure from a completely linear correlation. Notably, when eccentricity and angle values exceed approximately 1.120 and 75°, respectively, there is

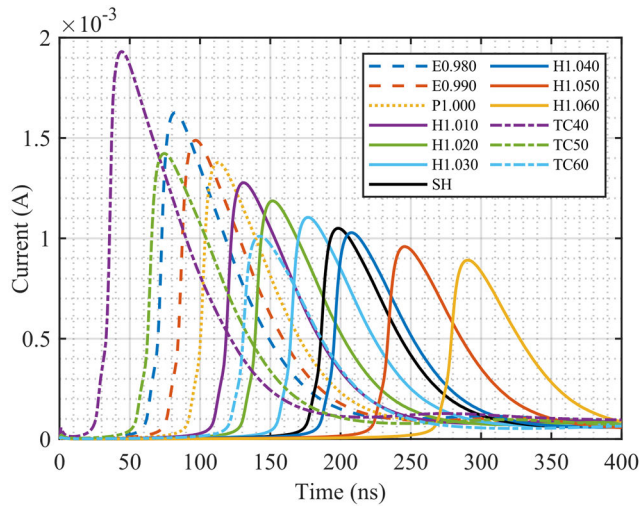


FIGURE 3. First Trichel current pulses for selected needle cases.

a noticeable expansion in the shapes of hyperbolic and tangent-circular needles. This expansion in electrode shape significantly reduces electric field non-uniformity near the needle tip, resulting in delayed or even absent characteristic discharge current pulsations, thus requiring higher voltage for their induction. Conversely, a decrease in eccentricity and angle values causes ellipsoid and tangent-circular electrodes to gradually approach a spherical and small hemispherically capped rod electrode shape, respectively. These shapes necessitate lower applied voltage levels to sustain pre-breakdown regimes and discharge current pulsations. Therefore, for meaningful comparisons under a uniform applied voltage of -5.5 kV, this study focuses exclusively on the mentioned eccentricity and angle ranges. To enhance organization, hyperbolic, parabolic, elliptic, and tangent-circular electrodes are identified using the initials H, P, E, and TC, along with their respective eccentricity or angle (e.g., H1.010 and TC40). In contrast, the base case is singularly marked with the abbreviation SH (standard hyperbolic) which is equal to H1.037.

Figure 3 illustrates the first Trichel pulse for selected cases, revealing two distinct patterns as the needle widens, corresponding to increased eccentricity or angle:

- 1) a reduction in discharge current magnitude, and
- 2) a delayed pulse onset.

Both patterns originate due to an intensified concentration of the electric field near the needle electrode tip as its body becomes narrower. For conic cross-sectional needle cathodes, these trends hold consistently across different geometric conic section shapes, with SH and P1.00 cases aligning with these trends. Yet, the inherent progression of eccentricity values suggests an order in terms of higher Trichel pulse magnitudes and frequencies: starting with elliptic, followed by parabolic, and culminating in hyperbolic configurations. These trends are further explored in Figures 4 and 5 for both needle configurations. Importantly, although presented together in unified figures, the data trends and relationships

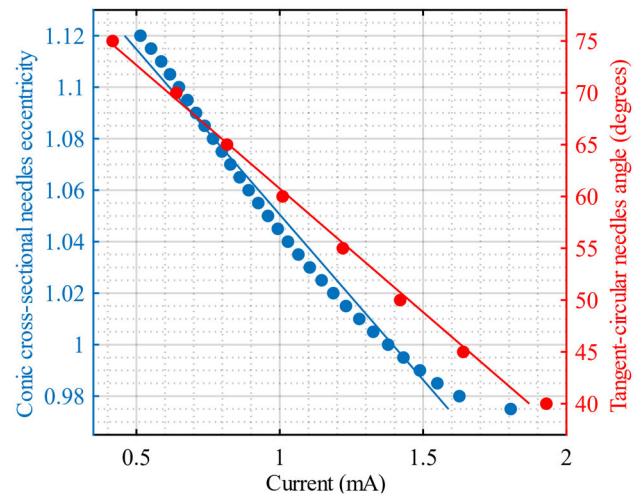


FIGURE 4. Peak magnitude of the first Trichel pulses for conic cross-sectional and tangent-circular needle cathodes and regression analysis.

are independent for conic cross-sectional and tangent-circular needles, with angle and eccentricity values being unrelated to each other.

In Figure 4, the peak magnitudes of the first Trichel pulse for all cases are depicted. Blue and red data points represent peak current magnitudes for conic cross-sectional and tangent-circular needles, respectively. Both needle types exhibit an almost linear inverse proportionality to eccentricity and angle changes, deviating slightly at the extremes of the studied range. Notably, there's a 72% and 78% decrease in peak current magnitude for conic cross-sectional and tangent-circular needles, respectively, as the needles widen within the entire studied range. Comparing elliptic and hyperbolic needles at the extremities of the eccentricity range, a 0.025 increase in elliptic eccentricity results in a 24% decrease in peak pulse magnitude and a 0.120 increase in hyperbolic eccentricity leads to a 63% decrease, despite sharing the same tip curvature. These contrasts underscore the inadequacy of solely relying on the geometric shape type and radius of curvature to define the shape of a conic cross-sectional needle electrode. The regression lines fitted to the data quantify these linear inverse relationships. Considering the slope of the regression lines, increases of 0.005 and 1° in eccentricity and angle values—corresponding to 0.091% and 0.097% increments in η , respectively—lead to decreases of 0.039 mA and 0.042 mA in the peak current magnitude of the first Trichel pulse for conic cross-sectional and tangent-circular needle cathodes, respectively.

Temporal attributes of Trichel current pulses are examined in Figure 5, showing occurrence times of the first pulse. These times include pulse onset and termination, corresponding to 10% of the peak current magnitude on rising and decaying edges, respectively, as well as the time of peak current magnitude. Evidently, there is a non-linear direct relationship between needle cathode eccentricity/angle values and pulse occurrence time. This non-linearity manifests as increasingly

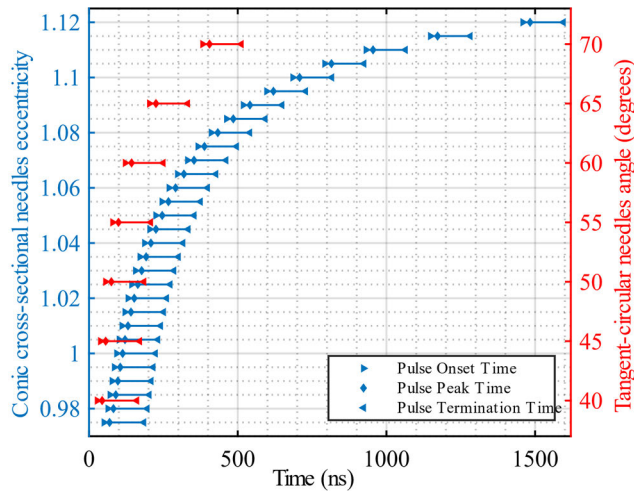


FIGURE 5. Onset, peak, and termination times of the first Trichel pulses for conic cross-sectional and tangent-circular needle cathodes.

greater delays in pulse occurrence as the needle cathode widens. This trend implies a higher frequency for Trichel pulse trains under narrower needle cathodes. Comparing the extremities of the studied ranges of eccentricity and angle values, the first pulse peak occurs 21.5 and 30.7 times later for conic cross-sectional and tangent-circular needle electrodes, respectively. These temporal instances exhibit nearly linear behavior under elliptic needles, with a 64% increase in pulse peak occurrence times per 0.025 increase in eccentricity—corresponding to 0.55% increase in η . However, for hyperbolic needles, this behavior is substantially non-linear, with a considerable 1212% increase in pulse peak occurrence time per 0.120 increase in eccentricity over the entire range—corresponding to a rise of 2.13% in η . Such variations, once more, underscore the significance of accurately defining needle electrodes. Importantly, changes in pulse occurrence times do not significantly affect Trichel pulses' width, rise time, and decay time; the rise time experiences a slight gradual increase with increasing eccentricity/angle, while the decay time and total pulse duration first decrease and then increase, forming a very wide U-shaped curve.

An important point to consider is the contrast in Trichel current pulse characteristics under conic cross-sectional needle cases to their approximated tangent-circular counterparts. For example, even though TC40 may seem like the ideal option for approximating the base case needle electrode SH—based on criteria like overlapping hyperbola asymptotes and tangents or close intersection points with the upper artificial boundary—a more thorough analysis unveils a significant disparity; in the case of TC40, negative corona result in discharge current pulsations that occur 4.5 times faster than those observed in SH, reaching a peak magnitude roughly 84% larger than that of SH. This significant difference primarily arises from the smoothly curved body shape of conic cross-sectional needle electrodes when compared to tangent-circular needles, which form an uneven

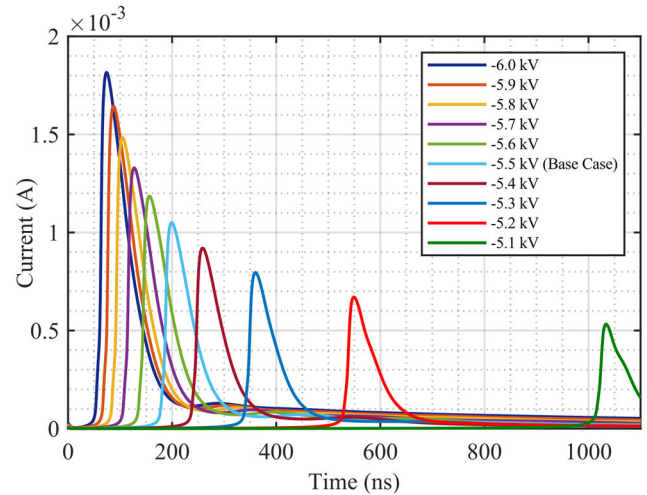


FIGURE 6. First Trichel current pulses under different applied voltages.

curvature with an obtuse angle between the circle and its tangents resulting in distinct electric field distributions. This finding challenges the feasibility of approximating a conic cross-sectional needle electrode using a tangent-circular needle with the same orientation and tip curvature.

D. IMPACT OF APPLIED VOLTAGE

As explained earlier, the applicable voltage range for having a pulsating discharge current encompasses values from the threshold voltage of Trichel pulses up to just before transitioning into glow discharge. Therefore, we examine an applied voltage range from -5.1 to -6.0 kV in 0.1 kV intervals, while keeping all other parameters consistent with the base case. Figure 6 illustrates the first Trichel current pulse across the range of investigated voltage levels. As previously documented (see e.g., [54]), two trends are visually discernible through these curves as the absolute value of the applied voltage decreases:

- 1) the discharge current magnitude diminishes, and
- 2) discharge current pulsation delays.

These trends can be attributed to the lower maximum electric field intensity observed in cases with lower absolute values of applied voltage. This reduced electric field intensity results in decreased particle mobility and slower rates of ionization, attachment, and secondary electron emission mechanisms. For a more comprehensive exploration, Figures 7 and 8 provide a detailed representation of these patterns. In Figure 7, the peak magnitude of the first Trichel pulse is displayed across all examined applied voltage levels. Here, a linear and direct relationship is evident between the absolute value of the applied negative DC voltage and the magnitude of the discharge current. This relationship is supported by a regression line fitted to the data. By analyzing the slope of this line, within the specified electrode system and simulation framework, we can anticipate an increase of 0.14 mA in the peak magnitude of the Trichel pulse for every 0.1 kV increase in the absolute value of the applied negative DC voltage.

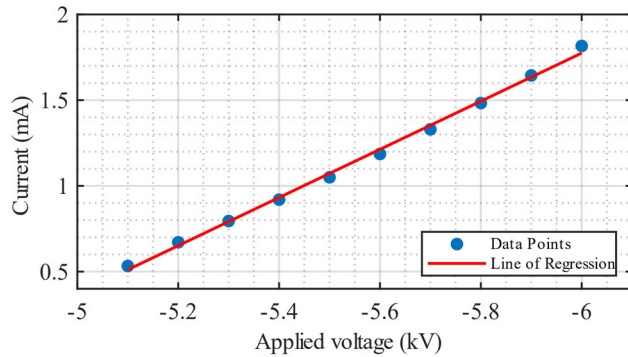


FIGURE 7. Peak Magnitude of the first Trichel pulses under different applied voltages and regression analysis.

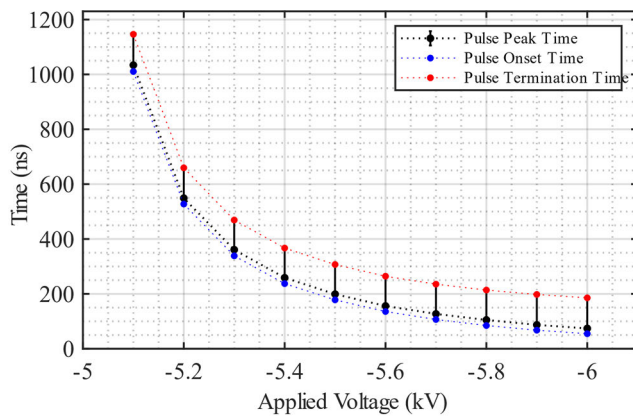


FIGURE 8. Onset, peak, and termination times of the first Trichel pulses under different applied voltages.

Figure 8 presents the onset, peak, and termination times of the first Trichel pulse under varying applied voltage levels within the studied spectrum. This observation suggests that the occurrence of pulsations in the discharge current undergoes a progressively larger temporal shift as the absolute value of the voltage decreases. This relationship is mathematically characterized by an inverse quadratic function, suggesting that as the absolute voltage decreases, the intervals between successive pulses lengthen, thereby reducing the frequency of the Trichel pulse train. Therefore, consistent with previous findings [17], [37], [54], the frequency of the Trichel pulses becomes directly proportional to the square of the absolute voltage, i.e., $f \propto U_a^2$. This quadratic proportionality means that any variation in voltage causes a disproportionately large change in the pulse frequency, demonstrating a non-linear and increasingly pronounced response in pulse occurrence times with decreasing absolute voltage values.

E. IMPACT OF DIFFERENT MESHING CONFIGURATIONS

In this section, we conduct a comparative analysis to explore the influence of different meshing configurations on the feasibility of reaching a converged solution, the accuracy and quality of simulation outputs, and the computational

resources and time needed for simulations. Our goal with such an investigation is to determine the minimum mesh resolution, with a reasonable margin of safety, necessary to obtain desired outputs in terms of accuracy, reproducibility, and visualization quality. This approach allows us to conserve valuable time and memory resources, which can be allocated to investigations involving multiple simulation scenarios. In this section, all simulations were conducted over a 400 ns study period, categorized as explored in the subsequent sections. The peak magnitude and the occurrence time of the first Trichel pulse are employed as the defining criteria and comparative benchmarks for evaluating the simulation output. The simulation parameters and results are elaborated in Table 2.

1) MESH MAXIMUM ELEMENT SIZE VARIATION (M1-M4)

In the first set of cases, we examine the effect of altering the maximum element size within the circular curve domain with a fixed radius of 0.1 mm. Specifically, we set these sizes at 0.0005 mm, 0.002 mm, 0.003 mm, and 0.004 mm for cases M1 to M4, respectively, compared to the base case's 0.001 mm. As expected, runtime and memory usage decrease with decreasing mesh resolution around the needle tip, following the order of M1, the base case, M2, and M3, while M4, featuring a coarser maximum mesh element size of 4 μm , fails to converge. This underscores the necessity of a minimum mesh resolution depending on the type of discharge plasma under simulation. In this category, both simulation runtime and memory usage demonstrate a direct near-linear proportionality with the total number of mesh elements. Despite significant differences in runtime, the simulation outputs remain remarkably consistent. However, as evident from Table 2, reduced mesh element size and resolution result in comparatively lower discharge current magnitudes, potentially indicating a decrease in reliability.

2) CIRCULAR CURVE (DOMAIN WITH FINER MESH) SIZE ADJUSTMENT (M5-M7)

The second group of cases focuses on assessing the impact of varying the radius of the circular area around the needle tip, while maintaining a maximum element size of 0.001 mm, consistent with the base case. Here, we set the radius to 0.01 mm, 0.05 mm, and 0.2 mm for cases M5, M6, and M7, respectively, while the base case had a radius of 0.1 mm. Expanding the coverage area with fine mesh increases the likelihood of effectively capturing and simulating main plasma activities. For instance, case M5, unlike other cases, featuring fine mesh coverage for an area with the smallest radius of 10 μm around the needle tip, did not converge. In terms of solution accuracy, a larger area with fine meshing provides more reliable results but demands higher time and memory resources. When comparing results of other cases to those of M7, which has the largest fine mesh coverage, it becomes clear that the base case, with larger coverage area compared to M6, is more reliable. Similar to the first category, in this category as well, simulation runtime and memory

TABLE 2. Simulation parameters and key results for mesh configuration study.

Case	Radius of circular curve domain with finer mesh (mm)	Maximum element size within circular curve (mm)	Total number of mesh elements	Simulation runtime (s)	Physical memory usage (GB)	Simulation output (the first Trichel pulse characteristics)	
						Peak magnitude (mA)	Peak occurrence time (ns)
Base case	0.1	0.001	53,673	2,593	4.54	1.050	199
M1	0.1	0.0005	129,441	6,611	9.23	1.051	198
M2	0.1	0.002	33,009	1,506	3.22	1.047	198
M3	0.1	0.003	28,588	1,279	3.02	1.043	198
M4	0.1	0.004	26,824	No Convergence			
M5	0.01	0.001	25,949	No Convergence			
M6	0.05	0.001	32,790	1,505	3.22	1.033	197
M7	0.2	0.001	138,439	6,755	9.29	1.050	198
M8	0.05	0.003	25,400	1,138	3.07	1.027	197
M9	0.2	0.0005	463,323	26,590	28.8	1.051	199

usage increase linearly with an increased number of mesh elements.

3) COMBINED MESH CONFIGURATION VARIATION (M8 AND M9)

The third category encapsulates studies involving circular regions with varying radii and meshing resolution, effectively combining the two aspects of the previous categories. As indicated in Table 2, cases M8 and M9 represent the lowest and highest combinations of circular curve radius and maximum element size (which yielded convergence in previous cases), respectively. As anticipated, case M8 resulted in the fewest total number of mesh elements, lowest simulation runtime, and memory used. However, simulation outputs in terms of result reliability and visualization quality were unsatisfactory. Conversely, case M9, with significantly higher runtime and memory usage, produced nearly identical results to the base case, rendering such a meshing configuration redundant. Given the results and analysis above, the base case offers a viable meshing configuration, striking a balance between ‘result accuracy, reliability, and visualization quality’ and ‘simulation runtime and required computational resources.

F. IMPACT OF DIFFERENT STABILIZATION TECHNIQUES

In this section, we engage in a comparative analysis to evaluate the influence of different stabilization techniques on the attainability of a converged solution, the accuracy, stability, and consistency of results, and the computational resources and time needed for simulations. Considering these factors, our goal is to determine the most appropriate technique for our specific case, i.e., negative corona discharges. It is important to emphasize that different discharge plasmas, depending on factors like electrode system, voltage levels, and dielectric material, may demand distinct stabilization techniques. In the following, we provide a comprehensive comparative analysis of different stabilization methods and their combinations. All simulations maintain a study duration of 400 ns, with the peak magnitude and occurrence time of

the first Trichel pulse serving as the criteria for comparing simulation outputs. Detailed simulation inputs and results are provided in Table 3.

1) CONSISTENT STABILIZATION (D1-D4)

First, we conducted simulations employing consistent anisotropic stabilization methods, both individually and in combinations. Across all these cases, the Trichel current pulses exhibited a degree of consistency regarding their magnitudes and frequencies (occurrence time). In general, Streamline diffusion, the technique used in our base case, demonstrated the shortest runtime, establishing itself as the most time-efficient approach. When considering the combined application of Streamline and Crosswind diffusions in cases D3 and D4, we observed a slightly longer runtime in comparison to Streamline alone. However, this approach resulted in a reduction in runtime when contrasted with the independent use of Crosswind methods (D1 and D2). Although the reduction may not be dramatic, it is indeed a noteworthy improvement. Additionally, in terms of Crosswind diffusion, we observed that the cases implementing Codina formulation—whether with or without Streamline diffusion—achieved faster convergence than the cases involving the Do Carmo and Galeão formulation; this trend is exemplified by D2 converging faster than D1, and D4 having a runtime shorter than D3. However, it is crucial to note that when considering computational resources, the Codina Crosswind, Streamline, and Do Carmo and Galeão Crosswind techniques showed a progressive increase in memory usage. Therefore, when comparing Streamline and Codina Crosswind diffusions, there is a trade-off between memory usage and runtime. On the other hand, despite all its advantages mentioned in Subsection II-F, the Crosswind method using the Do Carmo and Galeão formulation was found to be more memory-intensive and exhibited longer runtimes. When Streamline and Crosswind techniques are employed together, memory usage falls somewhere between the levels observed when each method is used in isolation. Although the differences in computational

TABLE 3. Simulation parameters and key results for different stabilization techniques.

Case	Stabilization technique				Simulation runtime (s)	Physical memory usage (GB)	Simulation output (the first Trichel pulse)	
	Anisotropic diffusion			Isotropic diffusion (δ_{id})			Peak magnitude (mA)	Peak occurrence time (ns)
	Streamline	Crosswind						
		Do Carmo and Galeão	Codina					
Base case	✓	-	-	-	2,593	4.54	1.050	199
D1	-	✓	-	-	3,341	4.67	1.054	200
D2	-	-	✓	-	2,673	4.38	1.051	199
D3	✓	✓	-	-	2,729	4.60	1.051	199
D4	✓	-	✓	-	2,655	4.47	1.050	199
D5	-	-	-	0.10	5,720	4.22	Unreliable Results	
D6	-	-	-	0.25	16,013	4.41	Unreliable Results	
D7	-	-	-	0.50	No Convergence			
D8	✓	-	-	0.10	16,405	4.54	Unreliable Results	

time and resource utilization may appear slight, they can become considerably more relevant when extending the study duration. Consequently, the choice between Streamline, Crosswind, or their combinations should be guided by the available memory and the desired simulation runtime.

2) INCONSISTENT STABILIZATION (D5-D8)

Next, we explored scenarios involving inconsistent isotropic diffusion introduced into the continuity PDEs. The introduction of such artificial diffusion with values of 0.10 (case D5) and 0.25 (case D6), and even their combination with anisotropic Streamline diffusion (case D8), led to highly noisy discharge current pulses with unrealistically high magnitudes. Cases D6 and D8 exhibit oscillations that even result in negative current values, which deviate from realism. Comparing cases D5 and D6 suggests that greater isotropic diffusion yields higher discharge current magnitudes but intensifies the scattering of noisy behavior. Following this, the introduction of even higher isotropic artificial diffusion with a value of 0.50 in case D7 leads to a complete lack of convergence. These results, marked by substantial noise and unphysical behavior, are not reported in terms of specific numerical values in Table 3. Notably, the simulation runtime for these cases is considerably longer compared to cases employing anisotropic diffusion, despite their comparatively lower memory usage. Interestingly, the combination of anisotropic diffusion with isotropic diffusion results in longer runtimes compared to using isotropic diffusion alone. It is crucial to reiterate that these findings may not necessarily apply universally to other types of gas discharge plasmas and should be assessed within the context of each specific simulation case.

G. IMPACT OF DIFFERENT LINEAR SYSTEM SOLVERS AND TIME-STEPPING METHODS

Given the general procedure of solving a set of PDEs outlined in subsection II-G, the selection of an appropriate

linear system solver and time-integration method assumes high importance. This subsection investigates the impact of various solvers and methods on the feasibility of reaching a converged solution, the accuracy of simulation outputs, and the computational resources and time needed for simulations. The primary goal of such a comparative study is to identify the optimal solver and settings tailored to the specific problem under consideration, ensuring desired outcomes in terms of convergence, accuracy, and efficiency. It is important to note that the conclusions drawn may not be universally applicable to all types of gas discharge plasma simulations. All simulations in this subsection have a consistent 400 ns study duration, facilitating meaningful comparisons. The criteria for evaluating simulation output are the peak magnitude and occurrence time of the first Trichel pulse. The results of these simulation cases are detailed in Table 4.

1) TIME-STEPPING METHODS AND SETTINGS (S1-S4)

All cases in this group employ *MUMPS* which is the default solver of COMSOL Multiphysics software for our model. The software categorizes time-stepping methods into two main types: *Implicit* (default) and *Explicit*, with the default time-stepping algorithm being the Backward Differentiation Formula (*BDF*). In case S1, BDF employs the highest possible *Minimum* and *Maximum orders*, set at 2 and 5, respectively, while the base case uses the least orders of 1 and 2. As anticipated, increased degrees of the interpolating polynomial of the BDF method, i.e., higher BDF orders, lead to longer runtimes and higher memory usage, as they store and operate on more previous time steps for higher-order accuracy. Additionally, as discussed in Section II, lower BDF orders are more effective in damping higher frequencies, thereby ensuring stability. While enhanced accuracy is anticipated with an increased BDF order, the results of S1 remain consistent with the base case. This suggests the redundancy of a higher-order BDF method for our specific discharge plasma under investigation. Thus, the choice of the proper BDF order

TABLE 4. Simulation parameters and key results for different time-stepping methods and linear system solvers.

Case	Solver	Time-Stepping Method		Simulation runtime (s)	Physical memory usage (GB)	Simulation output (the first Trichel pulse)	
		Implicit	Explicit			Peak magnitude (mA)	Peak occurrence time (ns)
Base Case	MUMPS	BDF (Max/Min BDF Order: 2/1)	-	2,593	4.54	1.050	199
S1	MUMPS	BDF (Max/Min BDF Order: 5/2)	-	2,753	4.71	1.050	199
S2	MUMPS	Generalized alpha	-	1,290	4.41	1.050	199
S3	MUMPS	Initialization only	-	-			
S4	MUMPS	-	Any method	No Convergence			
S5	PARDISO	BDF (Max/Min BDF Order: 2/1)	-	1,027	5.30	1.050	199
S6	SPOOLES	BDF (Max/Min BDF Order: 2/1)	-	Inordinate Solver Runtime and Minimal Progress			
S7	Dense matrix	BDF (Max/Min BDF Order: 2/1)	-	No Convergence			

should be made based on the specific gas discharge plasma phenomenon being simulated, considering the trade-offs between accuracy, stability, runtime, and memory usage. The subsequent case, S2, adopts the *Generalized alpha* as another *Implicit* time-stepping method. In comparison to *BDF* with a *Maximum order* of 2, *Generalized alpha* permits larger time steps, resulting in significantly less damping of high frequencies. This characteristic implies relatively higher accuracy but lower stability for *Generalized alpha* in contrast to second-order *BDF*. Therefore, *BDF* remains the preferred choice for handling stiff problems where additional robustness is crucial. However, *Generalized alpha* often proves to be an accurate method with satisfactory stability properties in many scenarios. In our specific case, for example, despite the presence of steep temporal variations and gradients in particle concentration—particularly during plasma sheath formation and around the occurrence time of discharge current peaks—the *Generalized alpha* method adeptly handled such stiffness, yielding results comparable to *BDF*. Additionally, as anticipated, the *Generalized alpha* method exhibited faster convergence and lower memory consumption. The *Initialization only* method, employed in the case of S3, enables performing a one-time initialization of the solution without advancing in time. This process involves applying initial conditions, solving for the initial state, and configuring the problem for transient simulation. This feature proves useful for verifying the correctness of initial conditions before proceeding with the time-dependent simulation. Moving on to S4, we assessed the performance of *Explicit* methods—both *Runge-Kutte* and *Adams-Bashforth* algorithms—with default settings, which failed to achieve convergence. Although explicit methods are computationally simpler to implement, they are not suitable for our stiff system of PDEs at hand.

2) LINEAR SYSTEM SOLVERS (S5-S7)

All cases in this category employ the *BDF* time integration method with a *Maximum order* of 2. This paper evaluates four available *Direct* linear system solvers of the software—*MUMPS*, *PARDISO*, *SPOOLES*, and *Dense matrix*—with their default settings. From Table 4, successful convergence with *MUMPS* and *PARDISO* underscores their compatibility and effectiveness in solving the hydrodynamic model for simulating our specific type of gas discharge plasma. Although the former achieved convergence with less memory usage, the latter converged sooner. The utilization of *SPOOLES* in S6 resulted in prolonged simulation with minimal progress, while *Dense Matrix* in S7 failed to converge. The struggle to converge with these two solvers implies their algorithmic incompatibility with the problem’s unique features and their inadequate handling of the matrix sparsity pattern and structural characterization.

IV. CONCLUSION

The first section of this paper has provided a comprehensive overview of the fundamental elements involved in simulating gas discharge plasmas using COMSOL Multiphysics software. It has provided the theoretical framework and software-related details of gas discharge plasma simulation components, with the aim of achieving meaningful outcomes. These components encompass the definition of parameters, variables, and functions used in the mathematical model, the representations of electrode geometries and computational domain, the establishment of initial and boundary conditions, the configuration of meshing, the application of stabilization techniques, and the fine-tuning of solver settings. A central theme throughout this paper was the pivotal role of correct and precise simulation inputs and the far-reaching implications of any inaccuracies.

The second section of this paper provides significant insights into negative corona discharges, emphasizing Trichel's current pulses. It was found needle electrode geometry critically influences discharge characteristics. Specifically, both the eccentricity and inter-tangent angle values of the conic cross-sectional and tangent-circular needle cathodes exhibited an indirect linear relationship with discharge current magnitude and an indirect quadratic relationship with the frequency of Trichel current pulse trains. Additionally, it was demonstrated that cross-sectional needle electrodes, representing the generic shape of needle electrodes, cannot be effectively approximated by their tangent-circular counterparts. Additionally, applied voltage variations significantly impact discharge current characteristics, such that increasing the absolute value of the applied voltage leads to an increase in both the magnitude and frequency of the Trichel current pulses.

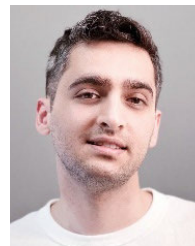
Our investigations showed finer meshing improves simulation accuracy, albeit increasing computational resource demands. Therefore, in adaptive meshing configurations, the area with fine mesh and its resolution should be optimized based on the specific type of plasma under study. Furthermore, the exploration of different stabilization techniques, linear system solvers, and time-stepping methods has been instrumental in improving the reliability and precision of our simulations. The case study illustrated that these settings should be carefully optimized considering the specific discharge plasma type under study.

These methodological enhancements have allowed for a more nuanced understanding of the dynamics of negative corona discharges. Overall, this paper not only advances the field of gas discharge plasma education and research but also provides a robust framework for future computational investigations, emphasizing the need for carefully chosen simulation parameters and techniques to achieve an optimal balance between accuracy and resource efficiency.

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